Math 122: Exam 2 - Practice 1

Student Name: _____

Instructions:

- Show all work: Full credit requires complete, correct steps and proper math notation.
- Points: Each question is worth 10 points.
- Formula Sheet: Permitted (formulas only; no worked examples).
- Calculators: Basic, non-graphing calculators allowed.

Part I: Integration by Parts

1. (10 points) Evaluate

$$I = \int x \ln x \, dx$$

2. (10 points) Evaluate

$$I = \int x^5 e^x \, dx$$

3. (10 points) Evaluate

$$I = \int \sin^{-1} x \, dx$$

4. (10 points) Evaluate the definite integral

$$\int_1^e x^2 \ln x \, dx$$

Part II: Trigonometric Integrals

5. (10 points) Evaluate

$$\int \sin^4 x \, dx$$

6. (10 points) Evaluate

$$I = \int \sin^4 x \cos^5 x \, dx$$

7. (10 points) Evaluate

$$I = \int \sin x \sin 2x \, dx$$

8. (10 points) Evaluate

$$\int 4 \sin x \cos 2x \sin 3x \, dx$$

Part III: Partial Fractions

9. (10 points) Evaluate

$$\int \frac{1}{(2x-1)(4x+2)}\,dx$$

10. (10 points) Evaluate

$$\int \frac{1}{(x-1)^2(x+1)}\,dx$$

End of Exam